FILE HIERARCHY STRUCTURE

Linux follows a tree-like directory structure, where all files and folders start from the root directory (/). Each directory has a specific role in system operation and file management.

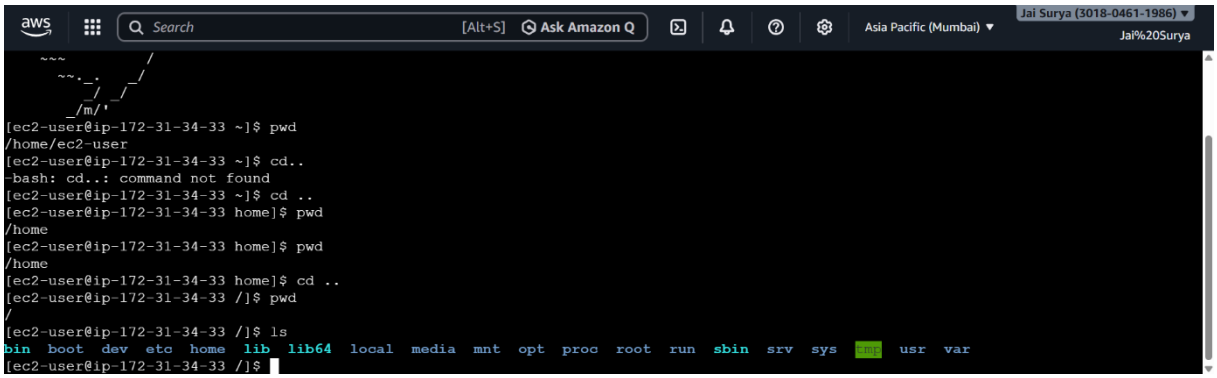
Important Linux Directories

- **/ –** Root directory (starting point of the file system)
- **/bin –** Essential user commands (ls, cp, mv)
- **/sbin** – System administration commands
- **/etc** – Configuration files
- **/home** – User home directories
- **/root** – Home directory of root user
- **/var** – Variable files (logs, mail, cache)
- **/tmp** – Temporary files
- **/usr** – User programs and utilities
- **/lib** – Shared libraries
- **/opt** – Optional application software
- **/dev** – Device files
- **/proc** – Process and kernel information

HANDS-ON FOR FILE HIERARCHY STRUCTURE

Screenshot: Linux File Hierarchy – Root Directory Navigation

1. Logged into the Linux terminal.
2. Executed pwd to check the current directory.
3. Used cd .. twice to navigate to the root directory (/).
4. Executed pwd to confirm the root location.
5. Used ls to list all top-level directories.



Screenshot: Navigating to /bin Directory

1. Executed cd /bin to move into the /bin directory.
2. Used pwd to confirm the current location as /bin.
3. Executed ls to list essential binary commands stored in /bin.

```
[ec2-user@ip-172-31-34-33 /]$ ls
bin  boot  dev  etc  home  lib  lib64  local  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
[ec2-user@ip-172-31-34-33 /]$ cd bin
[ec2-user@ip-172-31-34-33 bin]$ pwd
/bin
[ec2-user@ip-172-31-34-33 bin]$ ls
```

```
ac                  dnstap-read         jsonpatch-3.9       p11-kit             semodule_expand     timedatectl
acpi_listen         domainname          jsonpointer         package-cleanup     semodule_link       timeout
addr2line           dracut              jsonpointer-3       passwd              semodule_package    tload
alias               du                  jsonpointer-3.9     paste               semodule_unpackage  toe
amazon-ssm-agent    dumpkeys            jsonschema          pathchk             seq                 top
apropos             dwz                 kbd_mode            peekfd              sestatus            touch
apropos.man-db      ec2-metadata        kbdinfo             perl                set-hostname-imds   tput
ar                  echo                kbdrate             perl5.32.1          setarch             tr
arch                ed                  kernel-install      perldoc             setfacl             tracepath
arpaname            efisecdb            keyctl              pgrep               setfattr            traceroute
aspell              efivar              kill                pic                 setfont             traceroute6
as                  egrep               killall             piconv              setkeycodes         troff
at                  eject               kmod                pidof               setleds             true
atq                 elfedit             last                pidstat             setmetamode         truncate
atrm                encguess            lastb               pidwait             setpriv             trust
attr                env                 lastcomm            ping                setsid              tset
audit2allow         envsubst            lastlog             pinky               setterm             tsort
audit2why           eqn                 lchfn               pkg-config          setup-nsssysinit    tty
aulast              ex                  lchsh               pkgconf             setup-nsssysinit.sh tzbostat
aulastlog           expand              ld                  pkill               setup-policy-routes type
ausyscall           expr                ld.bfd              pldd                setvtrgb            tzselect
auvirt              factor              ld.so               pmap                sftp                uclampset
awk                 fallocate           ldd                 pod2man             sg                  udevadm
aws                 false               less                pod2text            sh                  ul
aws_completer       fc                  lessecho            pod2usage           sha1hmac            ulimit
b2sum               fg                  lesskey             portablectl         sha1sum             umask
base32              fgconsole           lesspipe.sh         post-grohtml        sha224hmac          umount
base64              fgrep               lexgrog             powernow-k8-decode  sha224sum           unalias
basename            file                link                pr                  sha256hmac          uname
basenc              fincore             linux-boot-prober   pre-grohtml         sha256sum           uname26
bash                find                linux32             preconv             sha384hmac          unexpand
```

Screenshot: Navigating to /boot Directory

1. Executed cd /boot to access the /boot directory.
2. Used pwd to confirm the current directory as /boot.
3. Executed ls to view boot-related files.

```
[ec2-user@ip-172-31-34-33 bin]$ cd /boot
[ec2-user@ip-172-31-34-33 boot]$ pwd
/boot
[ec2-user@ip-172-31-34-33 boot]$ ls
System.map-6.1.161-183.298.amzn2023.x86_64  efi    initramfs-6.1.161-183.298.amzn2023.x86_64.img  symvers-6.1.161-183.298.amzn2023.x86_64.gz
config-6.1.161-183.298.amzn2023.x86_64      grub2  loader                                         vmlinuz-6.1.161-183.298.amzn2023.x86_64
[ec2-user@ip-172-31-34-33 boot]$
```

Screenshot: Navigating to /dev Directory

1. Executed cd /dev to access the /dev directory.
2. Used pwd to confirm the current directory as /dev.
3. Executed ls to list device files.

```
[ec2-user@ip-172-31-34-33 boot]$
[ec2-user@ip-172-31-34-33 boot]$ cd /dev
[ec2-user@ip-172-31-34-33 dev]$ pwd
/dev
[ec2-user@ip-172-31-34-33 dev]$ ls
autofs          disk        log             loop7       nvme0n1p127  snapshot  tty12  tty20  tty29  tty37  tty45  tty53  tty61  ttyS3   vcs4  vcsa6  vhost-net
block           fd          loop-control    mapper      nvme0n1p128  stderr    tty13  tty21  tty3   tty38  tty46  tty54  tty62  uhid    vcs5  vcsu   vhost-vsock
btrfs-control   full        loop0           megaraid    nvram        stdin     tty14  tty22  tty30  tty39  tty47  tty55  tty63  uinput  vcs6  vcsu1  xvda
char            fuse        loop1           net         ptmx         stdout    tty15  tty23  tty31  tty4   tty48  tty56  tty7   urandom vcsa  vcsu2  xvda1
console         hpet        loop2           ng0n1       pts          tty       tty16  tty24  tty32  tty40  tty49  tty57  tty8   userfaultfd vcsa1 vcsu3  xvda127
core            hugepages   loop3           null        random       tty0      tty17  tty25  tty33  tty41  tty5   tty58  tty9   vcs     vcsa2 vcsu4  xvda128
cpu             initctl     loop4           nvme0       rtc          tty1      tty18  tty26  tty34  tty42  tty50  tty59  ttyS0  vcs1    vcsa3 vcsu5  zero
cpu_dma_latency input       loop5           nvme0n1     rtc0         tty10     tty19  tty27  tty35  tty43  tty51  tty6   ttyS1  vcs2    vcsa4 vcsu6
cum             kmsg        loop6           nvme0n1p1   shm          tty11     tty2   tty28  tty36  tty44  tty52  tty60  ttyS2  vcs3    vcsa5 vfio
[ec2-user@ip-172-31-34-33 dev]$
```

Screenshot: Navigating to /etc Directory

1. Executed cd /etc to access the /etc directory.
2. Used pwd to confirm the current directory as /etc.
3. Executed ls to view system configuration files.

```
[ec2-user@ip-172-31-34-33 dev]$
[ec2-user@ip-172-31-34-33 dev]$ cd /etc
[ec2-user@ip-172-31-34-33 etc]$ pwd
/etc
[ec2-user@ip-172-31-34-33 etc]$ ls
DIR_COLORS                chrony.conf      dracut.conf.d  hibinit-config.cfg  libibverbs.d    nanorc          profile.d         screenrc        sudoers.d
DIR_COLORS.lightbgcolor   chrony.d         environment    host.conf           libnl           netconfig       protocols         security        sysconfig
GREP_COLORS               chrony.keys      ethertypes     hostname            libreport       networks        rc.d              selinux         sysctl.conf
NetworkManager            cifs-utils       exports        hosts               libuser.conf    nfs.conf        rc0.d             services        sysctl.d
X11                       cloud            exports.d      idmapd.conf         locale.conf     nfsmount.conf   rc1.d             sestatus.conf   system-release
acpi                      cron.daily       filesystems    image-id            localtime       nsswitch.conf   rc2.d             shadow          system-release-cpe
adjtime                   cron.hourly      fstab          init.d              login.defs      openldap        rc3.d             shadow-         systemd
aliases                   cron.monthly     gcrypt         inittab             logrotate.conf  opt             rc4.d             shells          terminfo
alternatives              cron.weekly      gnupg          inputrc             logrotate.d     os-release      rc5.d             skel            tmpfiles.d
amazon                    crontab          groff          issue               lsm             pam.d           rc6.d             ssh             trusted-key.key
amazon-linux-release      crypto-policies  group          issue.net           machine-id      passwd          request-key.conf  ssl             udev
amazon-linux-release-cpe  csh.cshrc        group-         kernel              magic           passwd-         request-key.d     sssd            update-motd.d
at.deny                   csh.login        grub.d         keyutils            man_db.conf     pkcs11          resolv.conf       statetab.d      vimrc
audit                     dbus-1           grub2-efi.cfg  krb5.conf           mke2fs.conf     pkgconfig       rpc               subgid          virc
bash_completion.d         default          grub2.cfg      krb5.conf.d         modprobe.d      pki             rpm               subgid-         wgetrc
bashrc                    depmod.d         gshadow        ld.so.cache         modules-load.d  pm              rsyncd.conf       subuid          xattr.conf
bindresvport.blacklist    dhcp             gshadow-       ld.so.conf          motd            popt.d          rsyslog.d         subuid-         xdg
binfmt.d                  dnf              gss            ld.so.conf.d        motd.d          printcap        rwtab.d           sudo.conf       yum.repos.d
chkconfig.d               dracut.conf      gssproxy       libaudit.conf       mtab            profile         sasl2             sudoers
[ec2-user@ip-172-31-34-33 etc]$
```

Screenshot: Navigating to /home Directory

1. Executed cd /home to access the /home directory.
2. Used pwd to confirm the current directory as /home.
3. Executed ls to list user home directories.

```
[ec2-user@ip-172-31-34-33 etc]$
[ec2-user@ip-172-31-34-33 etc]$ cd /home
[ec2-user@ip-172-31-34-33 home]$ pwd
/home
[ec2-user@ip-172-31-34-33 home]$ ls
ec2-user
[ec2-user@ip-172-31-34-33 home]$
```

Screenshot: Navigating to /lib Directory

1. Executed cd /lib to access the /lib directory.
2. Used pwd to confirm the current directory as /lib.
3. Executed ls to view shared library files.

```
[ec2-user@ip-172-31-34-33 home]$ cd /lib
[ec2-user@ip-172-31-34-33 lib]$ pwd
/lib
[ec2-user@ip-172-31-34-33 lib]$ ls
binfmt.d  environment.d  grub       kbd     modprobe.d      motd        pam.d      sysctl.d        system-release-cpe  tmpfiles.d
debug     firmware       issue      kernel  modules         motd.d      python3.9  sysimage        systemd             udev
dracut    games          issue.net  locale  modules-load.d  os-release  rpm        system-release  sysusers.d
[ec2-user@ip-172-31-34-33 lib]$
```

Screenshot: Navigating to /lib64 Directory

1. Executed cd /lib64 to access the /lib64 directory.
2. Used pwd to confirm the current directory as /lib64.
3. Executed ls to view 64-bit shared libraries.

```
[ec2-user@ip-172-31-34-33 lib]$ cd /lib64
[ec2-user@ip-172-31-34-33 lib64]$ pwd
/lib64
[ec2-user@ip-172-31-34-33 lib64]$ ls
X11                     libdl.so.2                  libhunspell-1.7.so.0        libns-9.18.33.so            libsoftokn3.so
audit                   libdnf                      libhunspell-1.7.so.0.0.1    libnspr4.so                 libsolv.so.1
bfd-plugins             libdnf.so.2                 libibverbs                  libnss3.so                  libsolvext.so.1
bpf                     libdns-9.18.33.so           libibverbs.so.1             libnss_compat.so.2          libsqlite3.so.0
cifs-utils              libdrop_ambient.so.0        libibverbs.so.1.14.48.0     libnss_dns.so.2             libsqlite3.so.0.8.6
cracklib_dict.hwm       libdrop_ambient.so.0.0.0    libidn2.so.0                libnss_files.so.2           libss.so.2
cracklib_dict.pwd       libdw-0.188.so              libidn2.so.0.3.7            libnss_myhostname.so.2      libss.so.2.0
cracklib_dict.pwi       libdw.so.1                  libini_config.so.5          libnss_resolve.so.2         libssl.so.3
cryptsetup              libe2p.so.2                 libini_config.so.5.2.1      libnss_sss.so.2             libssl.so.3.2.2
dyninst                 libe2p.so.2.3               libinih.so.0                libnss_systemd.so.2         libssl3.so
engines-3               libeconf.so.0               libirml.so.1                libnssckbi.so               libsss_certmap.so.0
fipscheck               libeconf.so.0.7.9           libirs-9.18.33.so           libnsssysinit.so            libsss_certmap.so.0.2.0
games                   libedit.so.0                libisc-9.18.33.so           libnssutil3.so              libsss_idmap.so.0
gawk                    libedit.so.0.0.67           libisccc-9.18.33.so         libnuma.so.1                libsss_idmap.so.0.5.1
gconv                   libefa.so.1                 libisccfg-9.18.33.so        libnuma.so.1.0.0            libsss_nss_idmap.so.0
gettext                 libefa.so.1.2.48.0          libjansson.so.4             libonig.so.5                libsss_nss_idmap.so.0.6.0
gio                     libefiboot.so.1             libjansson.so.4.14.0        libonig.so.5.2.0            libsss_sudo.so
girepository-1.0        libefiboot.so.1.38          libjemalloc.so.2            libopcodes-2.41-50.amzn2023.0.5.so  libstdc++.so.6
groff                   libefisec.so.1              libjitterentropy.so.3       libpl1-kit.so.0             libstdc++.so.6.0.33
gssproxy                libefisec.so.1.38           libjitterentropy.so.3.4.1   libpl1-kit.so.0.3.0         libstoragemgmt.so.1
hmaccalc                libefivar.so.1              libjq.so.1                  libpl1.so.3                 libstoragemgmt.so.1.9.4
krb5                    libefivar.so.1.38           libjq.so.1.0.4              libpl1.so.3.5.0             libsystemd.so.0
```

Screenshot: Navigating to /media Directory

1. Executed cd /media to access the /media directory.
2. Used pwd to confirm the current directory as /media.
3. Executed ls to view mounted removable media.

```
[ec2-user@ip-172-31-34-33 local]$ cd /media
[ec2-user@ip-172-31-34-33 media]$ pwd
/media
[ec2-user@ip-172-31-34-33 media]$ ls
[ec2-user@ip-172-31-34-33 media]$
```

Screenshot: Navigating to /mnt Directory

1. Executed cd /mnt
2. Used pwd to confirm the directory as /mnt
3. Executed ls to view mount points

```
[ec2-user@ip-172-31-34-33 media]$
[ec2-user@ip-172-31-34-33 media]$ cd /mnt
[ec2-user@ip-172-31-34-33 mnt]$ pwd
/mnt
[ec2-user@ip-172-31-34-33 mnt]$ ls
```

Screenshot: Navigating to /opt Directory

1. Executed cd /opt to access the /opt directory.
2. Used pwd to confirm the current directory as /opt.
3. Executed ls to view optional application software.

```
[ec2-user@ip-172-31-34-33 lib]$ cd /opt
[ec2-user@ip-172-31-34-33 opt]$ pwd
/opt
[ec2-user@ip-172-31-34-33 opt]$ ls
aws
[ec2-user@ip-172-31-34-33 opt]$
```

Screenshot: Navigating to /proc Directory

1. Executed cd /proc to access the /proc directory.
2. Used pwd to confirm the current directory as /proc.
3. Executed ls to view process and kernel information.

```
[ec2-user@ip-172-31-34-33 opt]$
[ec2-user@ip-172-31-34-33 opt]$ cd /proc
[ec2-user@ip-172-31-34-33 proc]$ pwd
/proc
[ec2-user@ip-172-31-34-33 proc]$ ls
1     125   14    1435  1572  18    20    2545  28  36  44  773  8          cpuinfo        filesystems  key-users      locks     pagetypeinfo  swaps          version
10    1263  1402  1436  1587  183   2002  2597  29  37  5   774  829        crypto         fs           keys           mdstat    partitions    sys            vmallocinf
o
11    1264  1405  1440  1592  184   2003  2598  3   38  6   775  acpi       devices        interrupts   kmsg           meminfo   pressure      sysrq-trigger  vmstat
12    1265  1406  15    16    19    21    2599  30  4   67  776  buddyinfo  diskstats      iomem        kpagecgroup    misc      schedstat     sysvipc        zoneinfo
121   1266  1409  152   1602  1988  2282  26    32  40  70  777  bus        dma            ioports      kpagecount     modules   self          thread-self
1227  13    1410  154   1646  1993  23    2600  33  41  71  778  cgroups    driver         irq          kpageflags     mounts    slabinfo      timer_list
123   1302  1432  155   1703  1995  2385  27    34  42  74  779  cmdline    dynamic_debug  kallsyms     latency_stats  mtrr      softirqs      tty
1239  132   1433  1568  173   2     2542  2710  35  43  77  780  consoles   execdomains    kcore        loadavg        net       stat          uptime
[ec2-user@ip-172-31-34-33 proc]$
```

Screenshot: Navigating to /run Directory

1. Executed cd /run to access the **/run** directory.
2. Used pwd to confirm the current directory as **/run**.
3. Executed ls to view runtime system files.

```
[ec2-user@ip-172-31-34-33 proc]$
[ec2-user@ip-172-31-34-33 proc]$ cd /run
[ec2-user@ip-172-31-34-33 run]$ pwd
/run
[ec2-user@ip-172-31-34-33 run]$ ls
acpid.socket          auditd.pid  cloud-init   dbus                  gssproxy.sock  lock        motd     screen         sshd.pid    udev
agetty.reload         blkid       console      faillock              initctl        log         motd.d   sepermit       sudo        user
amazon-ec2-net-utils  chrony      credentials  gssproxy.default.sock initramfs      lsm         mount    setrans        systemd     utmp
atd.pid               chrony.d    cryptsetup   gssproxy.pid          irqbalance     machine-id  rpcbind  sm-notify.pid  tmpfiles.d
[ec2-user@ip-172-31-34-33 run]$
```

Screenshot: Navigating to /sbin Directory

1. Executed cd /sbin to access the **/sbin** directory.
2. Used pwd to confirm the current directory as **/sbin**.
3. Executed ls to view system administration commands.

```
[ec2-user@ip-172-31-34-33 run]$
[ec2-user@ip-172-31-34-33 run]$ cd /sbin
[ec2-user@ip-172-31-34-33 sbin]$ pwd
/sbin
[ec2-user@ip-172-31-34-33 sbin]$ ls
accessdb              debugfs       gdisk                    lgroupmod        nfsref                rtmon                                      tracepath6
accton                delpart       genhomedircon            lid              nfsstat               rtstat                                     tsig-keygen
acpid                 depmod        genhostid                lnewusers        nologin               runlevel                                   tune2fs
addpart               devlink       genl                     lnstat           nstat                 runuser                                    udevadm
adduser               dmfilemapd    getcap                   load_policy      ntsysv                sa                                         umount.nfs
agetty                dmsetup       getenforce               logrotate        pam_console_apply     sasldblistusers2                           umount.nfs4
alternatives          dmstats       getpcaps                 logsave          pam_namespace_helper  saslpasswd2                                unix_chkpwd
amazon-chrony-config  dosfsck       getsebool                losetup          pam_timestamp_check   sefcontext_compile                         unix_update
amazon-linux-sb       dosfslabel    groupadd                 lpasswd          parted                selabel_digest                             update-alternatives
arp                   dump-acct     groupdel                 lsmod            partprobe             selabel_get_digests_all_partial_matches    update-motd
arping                dump-utmp     groupmems                lspci            partx                 selabel_lookup                             update-pciids
atd                   dumpe2fs      groupmod                 luseradd         pidof                 selabel_lookup_best_match                  useradd
atrun                 e2freefrag    grpck                    luserdel         ping                  selabel_partial_match                      userdel
auditctl              e2fsck        grpconv                  lusermod         ping6                 selinux_check_access                       usermod
auditd                e2image       grpunconv                matchpathcon     pivot_root            selinuxconlist                             validatetrans
augenrules            e2label       grub2-bios-setup         mii-diag         plipconfig            selinuxdefcon                              vdpa
aureport              e2mmpstatus   grub2-get-kernel-settings mii-tool        poweroff              selinuxenabled                             vigr
ausearch              e2undo        grub2-install            mkdict           pwck                  selinuxexeccon                             vipw
autrace               e4crypt       grub2-mkconfig           mkdosfs          pwconv                semanage                                   visudo
avcstat               e4defrag      grub2-probe              mke2fs           pwhistory_helper      semodule                                   wipefs
badblocks             ebsnvme       grub2-reboot             mkfs             pwunconv              service                                    xfs_admin
```

Screenshot: Navigating to /srv Directory

1. Executed cd /srv to access the /srv directory.
2. Used pwd to confirm the current directory as /srv.
3. Executed ls to view service-related data.

```
[ec2-user@ip-172-31-34-33 sbin]$ cd /srv
[ec2-user@ip-172-31-34-33 srv]$ pwd
/srv
[ec2-user@ip-172-31-34-33 srv]$ ls
[ec2-user@ip-172-31-34-33 srv]$
```

Screenshot: Navigating to /sys Directory

1. Executed cd /sys to access the /sys directory.
2. Used pwd to confirm the current directory as /sys.
3. Executed ls to view kernel and system information.

```
[ec2-user@ip-172-31-34-33 srv]$
[ec2-user@ip-172-31-34-33 srv]$ cd /sys
[ec2-user@ip-172-31-34-33 sys]$ ls
block  bus  class  dev  devices  firmware  fs  hypervisor  kernel  module  power
[ec2-user@ip-172-31-34-33 sys]$
```

Screenshot: Navigating to /tmp Directory

1. Executed cd /tmp to access the /tmp directory.
2. Used pwd to confirm the current directory as /tmp.
3. Executed ls to view temporary files.

```
[ec2-user@ip-172-31-34-33 sys]$
[ec2-user@ip-172-31-34-33 sys]$ cd /tmp
[ec2-user@ip-172-31-34-33 tmp]$ ls
systemd-private-8cdb54ea491141d88efb97f05928d14a-chronyd.service-uDV4fK          systemd-private-8cdb54ea491141d88efb97f05928d14a-systemd-logind.service-28MYfr
systemd-private-8cdb54ea491141d88efb97f05928d14a-dbus-broker.service-d9koYO      systemd-private-8cdb54ea491141d88efb97f05928d14a-systemd-resolved.service-Uki51P
systemd-private-8cdb54ea491141d88efb97f05928d14a-policy-routes@ens5.service-IpfDlA
[ec2-user@ip-172-31-34-33 tmp]$
```

Screenshot: Navigating to /usr Directory

1. Executed cd /usr to access the /usr directory.
2. Used pwd to confirm the current directory as /usr.
3. Executed ls to view user programs and utilities.

```
[ec2-user@ip-172-31-34-33 tmp]$ cd /usr
[ec2-user@ip-172-31-34-33 usr]$ ls
bin  games  include  lib  lib64  libexec  local  sbin  share  src  tmp
[ec2-user@ip-172-31-34-33 usr]$
```

Screenshot: Navigating to /var Directory

1. Executed cd /var to access the /var directory.
2. Used pwd to confirm the current directory as /var.
3. Executed ls to view variable data such as logs and cache files.

```
[ec2-user@ip-172-31-34-33 usr]$ cd /var
[ec2-user@ip-172-31-34-33 var]$ ls
account  adm  cache  db  empty  ftp  games  kerberos  lib  local  lock  log  mail  nis  opt  preserve  run  spool  tmp  yp
[ec2-user@ip-172-31-34-33 var]$
```

HANDS-ON FOR LINUX COMMANDS

Screenshot: User Navigation and Privilege Switching

1. pwd

   → Shows current directory as **/home**.

2. cd ec2-user

   → Navigates to the **ec2-user** home directory.

3. pwd

   → Confirms current directory.

4. whoami

   → Displays the current logged-in user.

5. sudo su

   → Switches to the root user.

6. whoami

   → Confirms user as **root**.

7. exit

   → Exits root user and returns to previous user.

8. pwd

   → Displays current directory after exiting root.

```
[ec2-user@ip-172-31-34-33 var]$ cd /home
[ec2-user@ip-172-31-34-33 home]$ pwd
/home
[ec2-user@ip-172-31-34-33 home]$ cd ec2-user
[ec2-user@ip-172-31-34-33 ~]$ pwd
/home/ec2-user
[ec2-user@ip-172-31-34-33 ~]$ whoami
ec2-user
[ec2-user@ip-172-31-34-33 ~]$ sudo su
[root@ip-172-31-34-33 ec2-user]# whoami
root
[root@ip-172-31-34-33 ec2-user]# exit
exit
[ec2-user@ip-172-31-34-33 ~]$ pwd
/home/ec2-user
[ec2-user@ip-172-31-34-33 ~]$
```

- **$** → Indicates a normal user in Linux
- **#** → Indicates the root (administrator) user
- **~** → Represents the home directory of the current user

Screenshot: Basic Linux Commands Execution

1. mkdir folder

   → Creates a new directory.

2. pwd

   → Displays the current working directory.

3. ls

   → Lists files and directories.

4. cd folder

   → Navigates into the created folder.

5. touch file

   → Creates an empty file.

6. ls

   → Confirms file creation.

7. nano file

   → Opens the file in text editor. Save: **Ctrl + X → Y → Enter**

8. cat file

   → Displays file content.

9. rm file

   → Deletes the file.

10. ls

    → Confirms file deletion.

11. cd ..

    → Moves back to parent directory.

12. rm -rf folder

    → Deletes the directory.

```
[ec2-user@ip-172-31-34-33 ~]$ mkdir folder
[ec2-user@ip-172-31-34-33 ~]$ pwd
/home/ec2-user
[ec2-user@ip-172-31-34-33 ~]$ ls
folder
[ec2-user@ip-172-31-34-33 ~]$ cd folder
[ec2-user@ip-172-31-34-33 folder]$ touch file
[ec2-user@ip-172-31-34-33 folder]$ ls
file
[ec2-user@ip-172-31-34-33 folder]$ nano file
[ec2-user@ip-172-31-34-33 folder]$ cat file
this is my file
[ec2-user@ip-172-31-34-33 folder]$ rm file
[ec2-user@ip-172-31-34-33 folder]$ ls
[ec2-user@ip-172-31-34-33 folder]$ cd ..
[ec2-user@ip-172-31-34-33 ~]$ ls
folder
[ec2-user@ip-172-31-34-33 ~]$ rm -rf folder
[ec2-user@ip-172-31-34-33 ~]$ ls
```

Screenshot: File Rename and Copy Operations

1. touch file2

   → Creates a new file.

2. ls

   → Lists files to confirm creation.

3. mv file2 myfile

   → Renames **file2** to **myfile**.

4. ls

   → Confirms file rename.

5. nano myfile

   → Opens the file for editing

   → Save: **Ctrl + X → Y → Enter**

6. cat myfile

   → Displays file content.

7. touch secondfile

   → Creates another file.

8. cp myfile secondfile

   → Copies content from **myfile** to **secondfile**.

9. cat secondfile

   → Displays copied content.

10. cat myfile

    → Verifies original file remains unchanged.

```
[ec2-user@ip-172-31-34-33 ~]$
[ec2-user@ip-172-31-34-33 ~]$ touch file2
[ec2-user@ip-172-31-34-33 ~]$ ls
file2
[ec2-user@ip-172-31-34-33 ~]$ pwd
/home/ec2-user
[ec2-user@ip-172-31-34-33 ~]$ mv file2 myfile
[ec2-user@ip-172-31-34-33 ~]$ ls
myfile
[ec2-user@ip-172-31-34-33 ~]$ nano myfile
[ec2-user@ip-172-31-34-33 ~]$ cat myfile
hello all
[ec2-user@ip-172-31-34-33 ~]$ touch secondfile
[ec2-user@ip-172-31-34-33 ~]$ cp myfile secondfile
[ec2-user@ip-172-31-34-33 ~]$ cat secondfile
hello all
[ec2-user@ip-172-31-34-33 ~]$ cat myfile
hello all
[ec2-user@ip-172-31-34-33 ~]$
```